Does Adaptive Weighting Improve Equity Market Stress Measurement? Evidence from a GARCH–HMM–EVT Composite Index

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Abstract

This paper investigates whether combining conditional volatility, filtered market-regime probabilities, and extreme-loss severity through regime-conditioned entropy weights improves equity-market stress measurement. An Adaptive Composite Market Stress Index (A-CMSI) is constructed for AAPL, MSFT, and SPY from ARIMA–conditional-variance estimates, forward-filtered five-state Gaussian hidden Markov probabilities, and rolling peaks-over-threshold generalized Pareto expected shortfall. An optional specification adds cross-asset correlation. A chronological expanding-window design uses 2,765 adjusted-price observations from 2015 through 2025, monthly model refits, and mature-label forecast calibration. Across 1,810 held-out 30-day volatility forecasts, Adaptive yields RMSE 0.10532, compared with 0.10457 for equal weighting and 0.10411 for the NoHMM ablation. The Adaptive–Equal Diebold–Mariano test does not reject equal predictive accuracy ($p = 0.217$). Adaptive achieves a future 30-day crisis ROC AUC of 0.658 and an after-close same-day crisis-detection F1 of 0.300 at an expanding 90th-percentile signal threshold; these are distinct evaluation tasks. Rolling EVT and wavelet diagnostics characterize tails and timescale dependence, while external-index comparisons remain exploratory because revised observations and assumed publication lags are used. The findings support a reproducible stress-measurement framework but do not establish that adaptive weighting outperforms simpler composites.

Keywords: equity-market stress; composite indicator; GARCH; hidden Markov model; peaks over threshold; expected shortfall; entropy weighting; walk-forward forecasting.

JEL Classification: C58, G01, C53, G17.

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1. Introduction

A composite financial stress index seeks to summarize changing market conditions in a small number of interpretable quantities. Neither daily returns nor conditional volatility alone completely characterize the severity of adverse market episodes: two days with comparable estimated volatility may have different downside tail risks, and transient negative returns do not necessarily signal a persistent crisis regime. The problem is especially important when an index is evaluated as a potential early-warning signal rather than a historical description of episodes already observed.

Composite financial-stress measurement has substantial precedent. The ECB Composite Indicator of Systemic Stress (CISS) aggregates market-specific measures through an explicitly correlation-sensitive construction [1]. More recent work, including FIMSIS for Slovenia, compares alternative aggregation and risk-assessment methods [2]. These studies motivate transparent benchmark comparisons and caution against assuming that additional model complexity improves detection or prediction.

The present study evaluates an Adaptive Composite Market Stress Index (A-CMSI) for a restricted U.S. equity universe comprising Apple (AAPL), Microsoft (MSFT), and the SPDR S&P 500 ETF Trust (SPY). The core index combines three separately estimated components: a conditional-volatility signal V, a filtered adverse-regime probability P, and an extreme-loss expected-shortfall signal T. Trailing Shannon entropy determines baseline activity weights, which are altered by predefined multipliers depending on the expected HMM risk rank. A four-component variant additionally incorporates mean rolling pairwise correlations C.

The main empirical question is not whether GARCH, HMM, or EVT methods are individually new, or whether an index can visually highlight the COVID-19 crash. Rather, it asks whether an adaptive combination yields useful

incremental information relative to equal weights, PCA weighting, entropy-only weighting, and ablations in which particular components are removed. In-sample visual correspondence, after-close crisis detection, and strictly prospective forecasts are kept conceptually separate.

2. Related literature and contribution

2.1 Conditional risk, regime switching, and extreme tails

Time-varying conditional variance is rooted in ARCH and GARCH modeling [3, 4], with asymmetric extensions such as EGARCH designed to accommodate leverage-type effects [5]. Latent-state approaches to economic time series offer a probabilistic description of changing regimes [6]. In market-risk applications these models may capture different aspects of stress, but latent-state posterior probabilities must be handled cautiously when used in historical backtests: a smoothed probability conditions on future observations and differs from the information available at the historical close.

Tail-risk estimation using peaks over threshold draws on extreme-order asymptotics [7]. McNeil and Frey [8] combine GARCH filtering with extreme-value estimates for conditional VaR and expected shortfall. Salhi et al. [9] explicitly combine HMM and EVT in a regime-switching VaR model. Consequently, the pairing of HMM and EVT is an established research direction; the distinction investigated here lies in using them as separable, normalized inputs to a dated composite index with a regime-conditioned weighting mechanism.

2.2 Stress-index aggregation and testable contribution

Entropy is an information-theoretic functional [10], but using normalized entropy directly as a weight is not equivalent to the conventional entropy-deficit weighting $1 - H$. Here, higher temporal entropy receives greater baseline weight, which favors indicators showing more distributed absolute activity across the trailing window rather than isolated spikes. This is a declared operational design choice, not an inherently optimal principle. PCA and equal-weight comparators are retained to determine whether the intended adaptivity adds value.

The contribution is therefore a reproducible implementation and chronological comparison of a specific V–P–T index rather than a new theorem. The study makes three distinctions explicit: (i) multiple stress dimensions are estimated separately; (ii) regime-dependent weight changes use only forward-filtered probabilities in held-out periods; and (iii) component ablations and paired comparisons are allowed to favor a simpler composite.

3. Data and preliminary analysis

The dataset comprises 2,765 dated adjusted-closing-price rows for AAPL, MSFT, and SPY, from 2 January 2015 through 30 December 2025. Daily log differences produce 2,764 returns. The canonical input is the dated merged file `Data/prices_clean.csv` in the project repository; undated historical exports are not used for cross-asset alignment. VIX observations are retained from the project's previously saved HMM feature data and are not backfilled into dates before availability.

For asset a and trading date t, define adjusted closing price $S_{a,t}$, log return $r_{a,t}$, and downside loss $L_{a,t}$ as

$$r_{a,t} = \ln\left(\frac{S_{a,t}}{S_{a,t-1}}\right), \qquad L_{a,t} = -r_{a,t}, \qquad a \in \mathcal{A} = \{\text{AAPL}, \text{MSFT}, \text{SPY}\}. \tag{1}$$

Preliminary descriptive outputs include rolling return moments, 21-day realized variance, 63-day pairwise correlations, price and return figures, ADF and KPSS tests, log-price Johansen cointegration tests [11], conditional error-correction regressions where admissible, and structural-break diagnostics. These are contextual analyses rather than features assumed to improve the predictive index. A separate LOESS reconstruction is descriptive only and is excluded from forecasting.

This is an equity-focused rather than a system-wide financial-stability dataset. AAPL and MSFT are constituents of SPY, creating economically overlapping exposures. Accordingly, the mean of the three asset indices cannot be treated as independent cross-market diversification, and broad claims about banking, funding, or sovereign-debt stress would exceed the evidence.

Notation used throughout

Table 1: Core mathematical notation. Asset, component, and hidden-state indices are deliberately distinguished.

Symbol	Meaning
$a \in \mathcal{A}, t$	Asset ($\lvert\mathcal{A}\rvert = 3$) and after-close trading-date index.
$c \in C, d$	Stress component $c \in \{V, P, T\}$, with $d = 3$ for the main model and $d = 4$ for Adaptive4.
$s, j \in \{1, \ldots, M\}$	HMM hidden-state indices; $M = 5$ in the baseline, distinct from d.
$S_{a,t}, r_{a,t}, L_{a,t}$	Adjusted closing price, log return, and downside loss of asset a.
$\widehat{\sigma}_{a,t}$	Model-implied conditional standard deviation from the ARIMA–variance stage.
$\pi_t(s), \rho_s$	Forward-filtered probability of state s and that state's training-derived risk rank in $[0, 1]$.
P_t, g_t	Probability of the highest-risk state and posterior expected risk rank, respectively.
$u_{a,t}, \xi_{a,t}, \beta_{a,t}$	Rolling EVT threshold, GPD shape, and GPD scale.
$Z_{c,a,t}$	Clipped trailing-normalized component value; $Z_{P,a,t}$ is the same across assets.
$H_{c,a,t}, e_{c,a,t}$	Normalized temporal entropy and direct-entropy baseline weight.
$m_c(g_t), w_{c,a,t}$	Regime multiplier and final floored simplex weight.
W, ℓ, λ, h	Entropy lookback $W = 63$; normalization/EVT lookback $\ell = 252$; weight floor $\lambda = 0.05$; forecast horizon $h = 30$.

4. Methods

4.1 Conditional variance and standardized volatility

The conditional mean is modeled by ARIMA$(p, 0, q)$, where $p, q \in \{0, 1, 2\}$. BIC selects candidate specifications on the initial training sample. To display the conditional-variance input unambiguously, a generic ARMA representation of the differenced-free return model is

$$r_{a,t} = \mu_a + \sum_{i=1}^{p} \phi_{a,i} r_{a,t-i} + \varepsilon_{a,t} + \sum_{j=1}^{q} \theta_{a,j} \varepsilon_{a,t-j}, \quad \varepsilon_{a,t} = \sigma_{a,t} z_{a,t}, \quad \mathbb{E}[z_{a,t}] = 0, \quad \mathrm{Var}(z_{a,t}) = 1. \tag{2}$$

A standard GARCH$(1, 1)$ reference specification is

$$\sigma^2_{a,t} = \omega_a + \alpha_a \varepsilon^2_{a,t-1} + \beta_a \sigma^2_{a,t-1}. \tag{3}$$

Equations (2)–(3) identify the mean–variance decomposition; (3) is *not* a claim that plain GARCH was selected for every asset. The implementation compares GARCH, GJR-GARCH, and EGARCH candidate variance equations with Gaussian or Student-t standardized innovations, and selects among them using initial-sample BIC and convergence. The chosen models are re-estimated at 21-trading-day intervals on expanding samples, with parameters held fixed during each following block. Conditional volatility is updated as observed returns become available.

The operational volatility feature is a trailing, *after-close* z-score. For any scalar series x_t, write $\bar{x}_{\ell,t}$ and $s_{x,\ell,t}$ for its sample mean and standard deviation over dates $t - \ell + 1, \ldots, t$ (including t). Then

$$Z_{V,a,t} = \mathrm{clip}\left(\frac{\widehat{\sigma}_{a,t} - \bar{\sigma}_{a,\ell,t}}{s_{\sigma,a,\ell,t}}, -3, 3 \right), \qquad \ell = 252. \tag{4}$$

The normalizations are unavailable until sufficient trailing history exists. Historical R notebooks fitted specific asset-level candidates; the canonical Python pipeline performs its own chronological specification selection and exports operational V. The legacy precomputed normalized series is not renormalized in the principal experiment.

4.2 Forward-filtered hidden Markov component

A full-covariance Gaussian hidden Markov model uses SPY returns, conditional volatility, rolling realized volatility, rolling skewness, SPY drawdown, volatility of volatility, VIX, and mean pairwise rolling correlation. Feature scaling and model estimation are performed using only preceding observations. The baseline uses $M = 5$ hidden states and three deterministic-seed likelihood restarts per refit; two- through six-state choices were inspected on initial training data rather than chosen from held-out performance.

Let $\mathbf{x}_t$ denote the multivariate HMM feature vector, $A_{js} = \Pr(Q_t = s \mid Q_{t-1} = j)$ the fitted transition probabilities,

and $f_s(\mathbf{x}_t)$ the fitted Gaussian emission density in state s. The *forward-filtered*, normalized recursion is

$$\pi_t(s) = \frac{f_s(\mathbf{x}_t) \sum_{j=1}^{M} \pi_{t-1}(j) A_{js}}{\sum_{k=1}^{M} f_k(\mathbf{x}_t) \sum_{j=1}^{M} \pi_{t-1}(j) A_{jk}}, \qquad s = 1, \ldots, M. \tag{5}$$

The initial distribution is the fitted start probability (or the propagated final training posterior at a new fitted block). Crucially, (5) conditions on observations through t and does not use backward-smoothed, future-informed posteriors.

States are relabeled at each refit by a *training-sample* ranking using volatility, VIX, and returns. Define $s_\star$ as the highest-risk state and $\rho_s \in [0, 1]$ as its ordered risk score, with the highest ranked state receiving $\rho_{s_\star} = 1$. Then

$$P_t = \pi_t(s_\star), \qquad g_t = \sum_{s=1}^{M} \rho_s \pi_t(s), \qquad Z_{P,a,t} = \mathrm{clip}\left(\frac{P_t - \bar{P}_{\ell,t}}{s_{P,\ell,t}}, -3, 3 \right). \tag{6}$$

P_t is a common SPY-centered market regime signal in all three asset composites; it is not an independent HMM estimated for each individual asset.

4.3 Peaks-over-threshold expected shortfall

Negative log returns $L_{a,t}$ define downside losses. Within each trailing $\ell = 252$-day window, the empirical 90th percentile of losses sets a threshold $u_{a,t}$. Positive exceedances $Y = L - u$ conditional on $L > u$ are fitted by a generalized Pareto distribution with shape $\xi_{a,t}$ and scale $\beta_{a,t} > 0$, provided at least 20 exceedances exist. Its survival function is

$$\Pr(Y > y \mid L > u) = \left(1 + \frac{\xi y}{\beta} \right)^{-1/\xi}, \quad y \geq 0, \quad 1 + \frac{\xi y}{\beta} > 0. \tag{7}$$

Here and in the next two equations, (u, ξ, β) are the fitted parameters for the asset and rolling date. Let $p_u = n_u/\ell$ be the empirical fraction of losses exceeding u. For a tail confidence level α with $1 - \alpha \leq p_u$, POT extrapolation gives

$$\mathrm{VaR}_\alpha = u + \frac{\beta}{\xi} \left[\left(\frac{p_u}{1 - \alpha} \right)^{\xi} - 1 \right], \qquad \xi \neq 0, \tag{8}$$

$$\mathrm{ES}_\alpha = \frac{\mathrm{VaR}_\alpha + \beta - \xi u}{1 - \xi}, \qquad \xi < 1. \tag{9}$$

At $\xi \to 0$ the quantile limit is $\mathrm{VaR}_\alpha = u + \beta \ln[p_u/(1 - \alpha)]$. The operational tail component is the clipped trailing z-score of rolling $\mathrm{ES}_{0.99}$:

$$Z_{T,a,t} = \mathrm{clip}\left(\frac{\widehat{\mathrm{ES}}_{0.99,a,t} - \overline{\widehat{\mathrm{ES}}}_{a,\ell,t}}{s_{\widehat{\mathrm{ES}},a,\ell,t}}, -3, 3 \right). \tag{10}$$

Invalid fits or cases with $\xi \geq 1$ use a recorded *empirical* ES99 fallback instead of arbitrary shape clipping. A full-sample static comparison of 90th-, 95th-, and 99th-percentile thresholds, plus mean-excess and Q–Q diagnostics, supports descriptive assessment; none of these retrospective diagnostic fits are used to score held-out dates.

4.4 Direct entropy weights and regime adjustment

Raw V, P, and T signals are brought onto comparable scales via the clipped trailing z-scores above. For each asset a and component c, temporal proportions within a moving window of length $W = 63$ are based on *absolute* normalized activity:

$$p_{c,a,k|t} = \frac{|Z_{c,a,k}|}{\sum_{j=t-W+1}^{t} |Z_{c,a,j}|}, \qquad k = t - W + 1, \ldots, t. \tag{11}$$

For a nonzero denominator, define $0 \ln 0 := 0$ and the normalized Shannon entropy

$$H_{c,a,t} = -\frac{1}{\ln W} \sum_{k=t-W+1}^{t} p_{c,a,k|t} \ln p_{c,a,k|t}, \qquad e_{c,a,t} = \frac{H_{c,a,t}}{\sum_{j \in C} H_{j,a,t}}. \tag{12}$$

This is intentionally *direct entropy* H, not entropy-deficit $1 - H$. Consistent with the implementation, a component with a completely zero activity window has $H = 0$; when *all* available component entropies are zero, baseline weights

are assigned uniformly. If the component index set is $C = \{V, P, T\}$, the posterior expected risk rank g_t determines the documented linear endpoint multipliers:

$$m_V(g_t) = 1.3 - 0.5g_t, \quad m_P(g_t) = 0.8 + 0.3g_t, \quad m_T(g_t) = 0.9 + 0.5g_t, \qquad 0 \le g_t \le 1. \tag{13}$$

The risk-regime coefficients are modeling hypotheses, not empirically optimized multipliers. Let $q_{c,a,t} = e_{c,a,t} m_c(g_t)$. The final weights apply an *affine simplex floor* $\lambda = 0.05$:

$$w_{c,a,t} = \lambda + (1 - d\lambda)\frac{q_{c,a,t}}{\sum_{j \in C} q_{j,a,t}}, \qquad \sum_{c \in C} w_{c,a,t} = 1, \quad w_{c,a,t} \ge \lambda. \tag{14}$$

This map is applied when all needed features and weights are defined, with $d = |C|$. Asset-level and portfolio composite indices are then

$$\text{A-CMSI}_{a,t} = \sum_{c \in C} w_{c,a,t} Z_{c,a,t}, \qquad \text{A-CMSI}_{\text{portfolio},t} = \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \text{A-CMSI}_{a,t}. \tag{15}$$

The three-component index is primary. *Adaptive4* adds the normalized 21-day mean of the three pairwise return correlations, $Z_{C,t}$, with $m_C(g_t) = 1$ and $d = 4$. High mean correlation indicates synchronous movement but does not, without conditioning on signs, establish joint downside returns or cointegration breakdown.

4.5 Benchmarks and multiscale description

The comparisons are Equal, PCA, Entropy-only, NoEVT, NoHMM, NoGARCH, and Adaptive4. PCA uses normalized absolute leading-component loadings fitted only on earlier observations. NoEVT removes the tail component; NoHMM removes both the HMM panic probability and its regime multiplier; and NoGARCH substitutes rolling realized volatility for fitted conditional volatility. The Entropy-only model still applies the documented minimum-weight mapping but without regime multipliers. The benchmark methods use their own eligible component counts rather than treating omitted components as zero.

Wavelet and smoothed coherence diagnostics are computed using Morlet continuous wavelets [12], with edge regions masked. These are descriptive and include no wavelet statistical-significance claim.

5. Chronological design and outcome definitions

Table 2: Core experiment settings recorded in the saved run manifest.

Parameter	Setting
Price sample	2015-01-02 to 2025-12-30; 2,765 rows
Initial model training	756 trading days
Model refitting	Every 21 trading days
EVT lookback / threshold	252 days / 90th percentile
Minimum GPD excesses	20
Core HMM states	5; full-covariance Gaussian
Entropy lookback / weight floor	63 days / 0.05
Future-risk horizon	30 trading days
Scored forecast dates	2018-09-05 to 2025-11-14
Paired block bootstrap	1,000 repetitions

At each eligible after-close date, index values use observations through that date and models estimated on strictly earlier dates. Each index variant is calibrated using standardized expanding ridge regression to the logarithm of the next 30-day annualized realized volatility. Let $h = 30$ and let $s(\cdot)$ denote the sample standard deviation with denominator $h - 1$; the SPY forward-volatility target is

$$Y_t^{\text{vol}} = \sqrt{252}\, s\big(r_{\text{SPY},t+1}, \ldots, r_{\text{SPY},t+h}\big), \qquad h = 30. \tag{16}$$

A ridge predictor is fitted to $\ln Y_t^{\text{vol}}$ using mature training labels, with predictions exponentiated to return to annualized-volatility units. Logistic calibration separately estimates (i) the probability of crisis occurrence in the next 30 days

and (ii) the probability of a model-derived regime-rank transition. At least 126 matured out-of-sample examples are required for calibration, and overlapping label windows must be fully observable before entering the training set.

The same-day after-close stress indicator marks an SPY return below −3% or a trailing 20-day simple price return below −10%:

$$Y_t^{\text{crisis}} = \mathbf{1}\left\{ r_{\text{SPY},t} < -0.03 \;\vee\; \frac{S_{\text{SPY},t}}{S_{\text{SPY},t-20}} - 1 < -0.10 \right\}. \tag{17}$$

Expanding historical 75th-, 90th-, and 95th-percentile index thresholds produce after-close detection signals; thresholds are computed using *preceding* index observations. Early-warning events mark first crossings of the −10% trailing-return criterion with a 20-day refractory separation. Warning windows are 5, 10, and 20 trading days before event dates, excluding the event day. Contemporaneous detection is not interpreted as forecasting the same day's loss.

The future-volatility task uses RMSE, MAE, and direction-of-change accuracy. In particular, for held-out scored observations $t = 1, \ldots, n$, define

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(Y_t^{\text{vol}} - \widehat{Y}_t^{\text{vol}})^2}, \qquad \text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|Y_t^{\text{vol}} - \widehat{Y}_t^{\text{vol}}|. \tag{18}$$

Future-event discrimination uses ROC AUC, Brier score, and F1 at a stated probability cutoff. Forecast-loss differences are assessed by the Diebold–Mariano (DM) test [13] using a Bartlett/HAC long-run-variance estimate with 29 lags and the small-sample correction of Harvey, Leybourne, and Newbold [14]. For Adaptive and benchmark squared forecast errors, the signed differential is $d_t = e_{A,t}^2 - e_{B,t}^2$; the reported two-sided statistic is of the form

$$\text{DM}_{\text{HLN}} = \frac{\bar{d}}{\sqrt{\widehat{\Omega}_d / n}} \sqrt{\frac{n + 1 - 2h + h(h-1)/n}{n}}, \qquad h = 30, \tag{19}$$

where $\widehat{\Omega}_d$ is the Bartlett-weighted HAC long-run variance through lag $h - 1$. Paired 30-day circular-block bootstrap assesses RMSE differences with 1,000 replications. The resulting p-values are exploratory and not adjusted for multiple comparisons.

6. Empirical results

6.1 Volatility forecasting and component ablation

Table 3: Held-out index-variant comparison on common eligible forecast dates, n = 1,810. Lower RMSE/MAE and higher direction accuracy/crisis ROC AUC are better.

Index	RMSE↓	MAE↓	Direction↑	Crisis AUC↑
NoHMM	**0.10411**	**0.05638**	**0.6785**	0.6573
Equal	0.10457	0.05681	0.6646	0.6776
Entropy	0.10478	0.05677	0.6696	0.6656
Adaptive4	0.10527	0.05762	0.6663	0.6611
PCA	0.10528	0.05703	0.6729	**0.6843**
Adaptive	0.10532	0.05692	0.6646	0.6583
NoGARCH	0.10578	0.05720	0.6519	0.6468
NoEVT	0.10678	0.05974	0.6696	0.5919

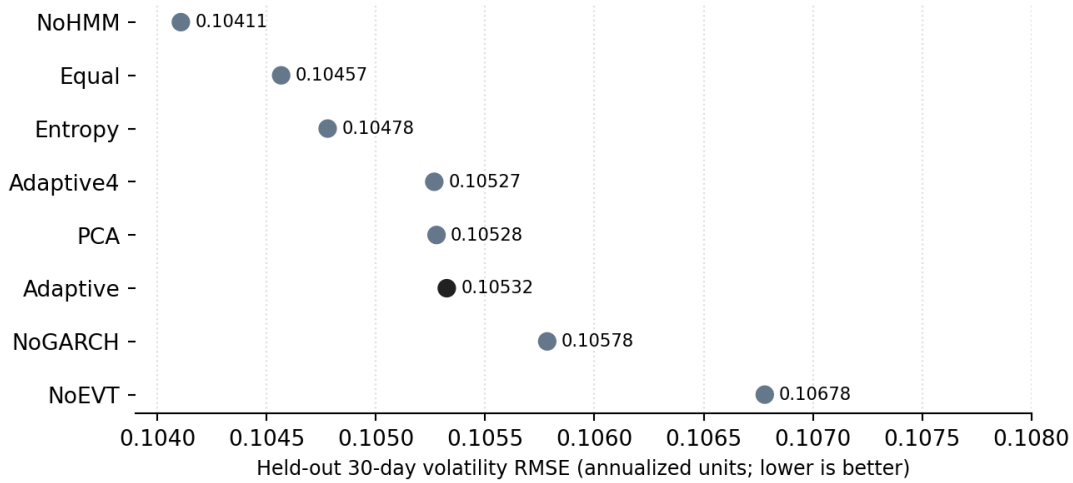


Figure 1: Comparative held-out RMSE of eight index variants. Axis is zoomed to show small differences. Reproduced from the repository's `main_ablation_metrics.csv`; original empirical figure retained.

NoHMM obtains the lowest annualized-volatility RMSE (0.10411), while Adaptive yields 0.10532 and Equal yields 0.10457. The model complexity of Adaptive is therefore not rewarded by lower squared forecast error in this study. Nonetheless, removal of EVT raises RMSE to 0.10678, and NoEVT also has a lower future-crisis ROC AUC of 0.5919 than Adaptive's 0.6583. These differences are not sufficient by themselves to establish causal or statistically significant contributions of the corresponding components.

6.2 Significance and stability of the weighting mechanism

Table 4: Exploratory paired forecast comparisons of Adaptive against selected baselines. Positive RMSE improvement means Adaptive performs better.

Comparator	DM stat.	p-value	RMSE improvement	95% bootstrap interval
Equal	1.2351	0.2169	−0.00076	[−0.00152, 0.00034]
NoHMM	0.9525	0.3410	−0.00122	[−0.00377, 0.00120]
NoEVT	−0.8694	0.3847	+0.00146	[−0.00165, 0.00574]
NoGARCH	−1.3427	0.1795	+0.00046	[−0.00018, 0.00121]

No reported interval in Table 4 excludes zero. The Adaptive–Equal DM test yields $p = 0.217$, so the data do not support a claim of superior volatility predictive accuracy from adaptive weighting. Sensitivity analysis with entropy lookbacks of 21, 63, and 126 trading days reports Adaptive RMSE approximately 0.1055, 0.1053, and 0.1053, respectively. The limited variation applies to the examined window choice only, not to state count, multipliers, asset composition, or crisis definitions.

6.3 Extreme-value and multiscale diagnostics

Table 5: Descriptive full-sample fitted GPD shape parameter $\widehat{\xi}$ by asset and loss threshold quantile.

Asset	90th percentile	95th percentile	99th percentile
AAPL	0.107	0.121	0.105
MSFT	0.100	0.201	0.244
SPY	0.184	0.232	0.304

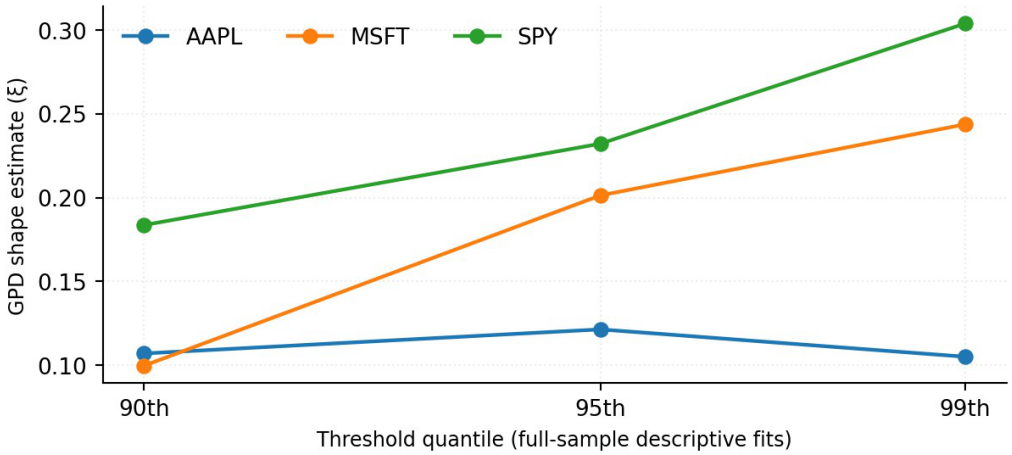


Figure 2: Fitted GPD shape estimates at selected *descriptive* thresholds. At the 99th-percentile threshold only 28 sample exceedances remain per asset; these are not precise tail constants. Original empirical figure retained.

Positive estimated shape parameters are consistent with heavier right tails for downside losses under the fitted model. AAPL is comparatively stable over the three static thresholds; estimates for MSFT and SPY vary more. The operational 252-day rolling 90th-percentile POT approach usually provides roughly 25 exceedances, introducing nontrivial estimation uncertainty in extrapolated 99% Expected Shortfall. One of 7,539 rolling tail fits required an empirical fallback, which was explicitly logged.

6.4 Crisis detection and early warnings

Table 6: Adaptive same-day crisis detection *after close*, n = 2,008. These are not same-day forecasts.

Index threshold	Precision	Recall	F1
75th percentile	0.0932	0.8596	0.1681
90th percentile	0.1967	0.6316	0.3000
95th percentile	0.3158	0.5263	0.3947

The selected expanding 90th-percentile decision rule yields after-close detection F1 of 0.300. Changing to the 95th percentile improves precision and F1 at the expense of recall. Such threshold results are contemporaneous assessments formed after observing the day's market return; they do not constitute an out-of-sample prediction of that same return. For separately calibrated 30-day future crises, Adaptive ROC AUC is 0.658, while PCA reaches 0.684 and Equal reaches 0.678 in the reported sample.

Table 7: Pre-event last threshold crossings, conditional on event coverage.

Event date	Last crossing	Lead (trading days)
2018-12-24	2018-12-17	5
2020-03-06	2020-02-25	8
2022-05-09	Not observed	—
2022-09-23	2022-09-14	7
2025-04-04	2025-03-11	18

Four of five eligible episodes have a preceding threshold crossing within the defined 60-trading-day search interval. The median across those four observed leads is 7.5 trading days; the fifth event does not count as a successful warning. With only five eligible episodes and possible false alarms outside the event windows, these figures remain exploratory.

6.5 External benchmark comparison

The optional comparison uses Federal Reserve Financial Stress Index series STLFSI4 and KCFSI with assumed observation-release delays of 7 and 45 calendar days, respectively. On the matched external comparison sample, an external-only forecasting model has RMSE 0.11556 and a model augmented with A-CMSI has RMSE 0.11065.

These figures are not directly ranked against Table 3 because the eligible sample differs. Moreover, the inputs are revised published histories rather than vintage-correct historical releases. The comparison is descriptive incremental evidence, not point-in-time proof of tradable foresight.

7. Interpretation and limitations

The main negative result is informative: a regime-conditioned entropy mechanism can deliver a plausible and visually interpretable stress index without improving the primary held-out volatility-loss metric. One possible explanation is redundant statistical information among conditional volatility, filtered panic probability, and expected shortfall. Another is that economics-inspired regime multipliers overfit the conceptual importance of an indicator without aligning with the chosen predictive loss. The present experiment does not identify the mechanism uniquely; it demonstrates that additional adaptivity must be empirically justified.

Several methodological qualifications delimit the interpretation. First, the risk universe is restricted to two large technology stocks and a broad U.S. equity ETF containing those stocks. Second, a 252-day 90th-percentile GPD window is sparse for stable far-tail estimation. Third, the five-state HMM and bull/panic multipliers are analyst choices whose sensitivity was not fully explored. Fourth, scores are operationally after-close and cannot establish intraday reaction or same-day prevention.

Fifth, future realized-volatility targets overlap, reducing the effective number of independent forecast observations. HAC and block bootstrap procedures mitigate some serial-dependence concerns but do not resolve uncertainty from exploratory comparison selection. Sixth, daily adjusted-price histories and external financial-stress series may incorporate revisions unavailable at the time. Seventh, wavelet coherence is interpreted without formal Monte Carlo significance contours. Eighth, the regime-transition target is a model-derived proxy rather than independently observed financial-crisis ground truth.

A natural extension would compare direct-entropy weights H with entropy-deficit weights $1 - H$, expand to non-overlapping asset groups and credit/funding indicators, evaluate alternative EVT windows and thresholds, obtain point-in-time external benchmark vintages, and replicate in new periods. Such work could show whether adaptive weighting has localized downturn benefits even when average volatility RMSE is unchanged.

8. Conclusion

A-CMSI provides a transparent, reproducible, after-close composite equity-market stress measurement experiment by integrating forward-operational conditional volatility, filtered latent-regime probabilities, and rolling EVT tail severity. Dynamic entropy weights and regime multipliers are tested against simpler aggregation and component-ablation alternatives under chronological forecast calibration. For 1,810 scored held-out forecasts, Adaptive has RMSE 0.10532 compared with 0.10457 for Equal and 0.10411 for NoHMM; its difference from Equal is not statistically significant in the reported test. A-CMSI retains exploratory value as a multidimensional stress summary, but the evidence does not support a universal forecasting advantage. Broader replication and more demanding tail-risk and point-in-time assessments should precede operational decision claims.

Statements and reproducibility

Code and data availability. The reproducible implementation, saved derived outputs, figures, and run manifest are available at `https://github.com/ayushanand1103/A-CMSI-research`. The primary command is `python run_acmsi.py`. Adjusted-price inputs and external stress series are third-party data subject to their original terms.

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Author responsibility. This is a sole-authored, independently conducted project, not undertaken on behalf of a university or other institution. The author is responsible for verifying the manuscript, external sources, data provenance, and disclosures before any submission. Version 2 revises presentation and adds explicit mathematical definitions; numerical results reported here are carried forward from the original preprint without recomputation.

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